

Engineering Design Process | XRP Bridge to *FIRST*®

3D Printing Guide

Module Information

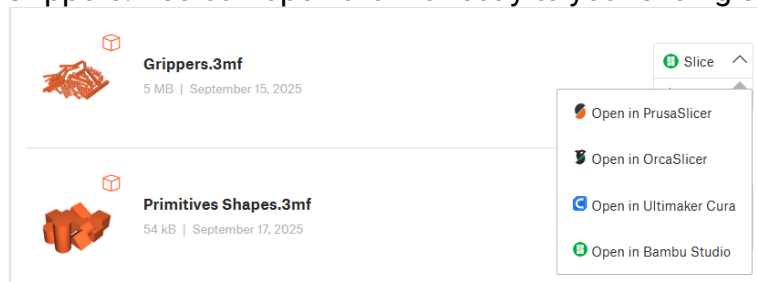
ABOUT THE 3D PRINTING GUIDE

- Items and print times were created using a Bambu P1P printer. They are suggested and may vary according to the 3D printer you are using.
- The 3D Printing Guides are created for one set of complete materials that can be used for a group of two students. You will want to multiply this for your needs.
- To access the files, click this link on [Printables](#)

1. Click on the Engineering Design folder:



2. This lesson has two files that are needed for the lesson, Primitives Shapes and Grippers. You can open them directly to your slicing software.



3. Alternatively download the files by clicking the link (they should automatically download).



4. If you use the download method, you will need to import the quantity of each file needed for the module to your slicing software.
5. Follow any suggestions for part orientation and supports for printing.
6. Slice the print to build the g-code for printing.
7. Prepare your 3D printer as needed.
8. Print.

Lesson 3 | 3D Primitives

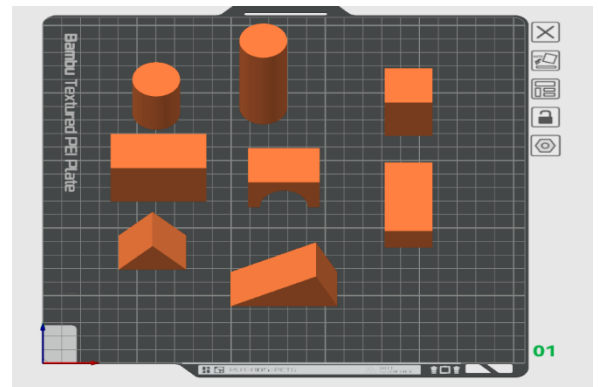
PARTS LIST

Following, you will find a list of parts for this lesson. You will want to clone the parts to equal the total quantity of each item on your build plate before printing. You can download the zip file of the parts by clicking the [Primitive STL Files](#). All items were created in Onshape, and the original files can be accessed from the [Primitive Shapes](#) folder.

Item Name	Quantity
Arch	1
Cube	1
Cylinder (Short)	1
Cylinder (Tall)	1
Rectangular Prism (Half)	1
Rectangular Prism	1
Triangular Prism (Right Isosceles)	1
Triangular Prism (Right)	1

PARTS ORIENTATION BUILD ARRANGEMENT

When printing the Primitive Shapes, you will want to ensure that the Rectangular Prism files and the Triangular Prism files are laid flat on the build surface to reduce printing time.



SUPPORTS AND PRINTING TIME

The table at the right shows estimated amounts of filament and print time using a Bambu P1P printer. If you wish to reduce print time, you can reduce sparse infill settings.

Total Estimation		
Total Filament:	18.41 m	55.81 g
Model Filament:	18.41 m	55.81 g
Cost:	1.39	
Prepare time:	5m46s	
Model printing time:	1h25m	
Total time:	1h31m	

Lesson 5 | Gripper Decisions

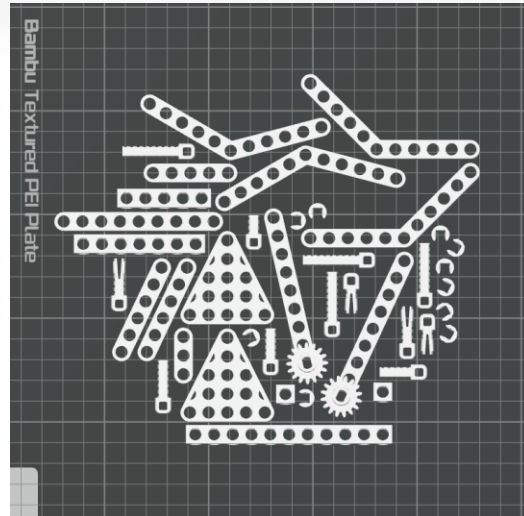
PARTS LIST

Following, you will find a list of parts for this lesson. You will want to clone the parts to equal the total quantity of each item on your build plate before printing. You can download the zip file of the parts by accessing Printables. All items were created in Onshape, and the original files can be accessed from the [XBS Gripper Decisions](#) folder.

Item Name	Builds			Total Quantity
	Stationary Gripper Quantity	Angular Gripper Quantity	Radial Gripper Quantity	
X-Shaft-5-T	1		1	2
X-Plate-1-9-6mm	1			1
X-Plate-Triangle-3-5-F	2			2
X-PlateAngled-6-6-13deg-6mm	2			2
X-Plate-1-5-6mm	1			1
X-Shaft-9-T	2			2
X-Clip	3		7	10
X-Shaft-B-T			2	2
X-Bar-7-1			1	1
X-Bar-11-1			1	1
X-Plate-1-3-3mm			1	1
X-Shaft-1-B-T			2	2
X-Plate-1-6-3mm			2	2
X-Shaft-3-T			1	1
X-Bar-1-1			2	2
X-Shaft-6-T			1	1
X-PlateAngled-6-6-13deg-3mm			2	2
X-GearSpur-16-6mm-3mm-D-B8		2		2
X-Shaft-3B-T		2		2
X-Bar-5-1		1		1

PARTS ORIENTATION AND SUPPORTS

The parts in this lesson can all be laid flat on the plate and do not need any additional support for printing. They are all printed using standard PLA.



PRINTING TIME, ESTIMATED COST, AND FILAMENT USAGE

This may vary depending on your printer but can give you the specifications that the files were printed during testing.

With a Bambu P1P, this is the estimated print time and filament usage.

Slicing Result		
Color Scheme ~ Filament		
Filament	Model	
■ 1	14.93 m	44.52 g
Filament change times: 0		
Cost: 0.89		
Time Estimation		
Prepare time:	6m27s	
Model printing time:	4h51m	
Total time:	4h57m	